

BALI: Solvency-Constrained Batch Liquidation as a Cryptoeconomic Control Mechanism

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Abstract—We study liquidation in onchain perpetual markets as a stochastic control problem with delayed marks and finite liquidity. We evaluate batch liquidation (BALI), a solvency-constrained mechanism with stress triggers, uniform-price clearing, concentration caps, and deterministic fallback, against a mathematically comparable continuous liquidation baseline. Results are conditional: BALI consistently reduces cascade depth, while impact and bad debt improve only when auction participation is sufficiently elastic and fallback activates before uncovered residuals approach bankruptcy. Under weaker participation or larger effective delay, continuous liquidation can perform better on impact or bad debt. These findings position batch liquidation (BALI) as a state-contingent control layer rather than a universal replacement.

Index Terms—Perpetual futures, liquidation, batch auction, mechanism design, systemic risk, DeFi market microstructure.

I. Introduction

Onchain perpetual markets concentrate risk in liquidation pipelines. When prices move quickly and depth thins, liquidation flow becomes mechanically pro-cyclical: forced selling (or buying) widens impact, degrades account equity on other books, and triggers additional liquidations. This feedback loop is now a first-order design problem rather than an edge case.

Current designs prioritize immediacy via continuous liquidation. This minimizes control-loop delay but does not internalize market-impact externalities generated by sequential forced executions. Conversely, purely delayed mechanisms can improve execution quality while increasing insolvency risk if shortfalls drift during waiting windows.

This paper asks: Can liquidation be redesigned as a solvency-constrained market-clearing layer that trades a small amount of latency for lower impact and lower cascade intensity, without relaxing hard solvency safeguards?

Our contributions are threefold:

- 1) We formalize liquidation as a control problem under latency and liquidity constraints, with explicit bad-debt, impact, and cascade objectives.
- 2) We propose BALI as a coordinated liquidation mechanism with precise trigger logic, uniform-price clearing, concentration-bounded allocation, and emergency fallback hierarchy.
- 3) We provide paired simulation evidence against a rigorously specified continuous liquidation baseline

across normal, volatility-spike, and correlated-shock regimes.

Our results suggest that the primary design axis in liquidation systems is the tradeoff between coordination and latency, rather than the specific auction format.

The key practical result is conditional: BALI reliably reduces cascade depth in our simulations, while impact and bad-debt performance depend on auction participation quality and fallback thresholds. Continuous liquidation remains a necessary fallback under participation failures, motivating state-contingent operation rather than one-size-fits-all liquidation policy.

II. Related Work

A. Batch-auction market design

Frequent batch auction (FBA) research shows that discrete-time clearing can reduce latency races and improve welfare under particular information structures [1], [2], [3]. However, these papers study general exchange execution and do not model margin-triggered forced deleveraging, bankruptcy boundaries, or bad-debt control.

B. ADL and liquidation research in perpetuals

Recent ADL papers formalize fairness–revenue–solvency trade-offs in loss socialization and queue optimization [4], [5]. Related protocol studies examine keeper incentives and liquidation throughput in collateral auctions [7]. This line focuses primarily on post-failure or protocol-specific liquidation tuning, not an integrated primary clearing mechanism that explicitly optimizes impact and cascade depth before ADL becomes active.

C. DeFi empirical and systemic-risk literature

DeFi systems papers characterize MEV externalities, liquidation races, and composability-driven contagion [8], [9], [10], [6]. These works provide empirical risk factors and diagnostics but stop short of specifying a full mechanism with trigger rules, allocation policy, and fallback solvency constraints.

D. Positioning and novelty

To our knowledge, this work is among the first to model liquidation as a control problem under latency and liquidity constraints in onchain perpetual markets, and to

instantiate that model as a deployable coordinated liquidation mechanism with explicit emergency fallback. Batch auctions represent one instantiation of a broader class of coordinated liquidation policies that trade off latency and impact. Concretely, unlike FBA exchange papers, we optimize solvency-linked objectives (bad debt, cascade depth) rather than generic execution welfare; unlike ADL papers, we target pre-ADL primary liquidation clearing; unlike empirical/system studies, we provide mechanism-level policy and implementation details. This separation is central to our contribution and to fair comparison across the broader coordinated-liquidation design space.

III. System Model

A. State and account model

Time is discrete at block index $t \in \{0, 1, \dots, T\}$. For market $m \in \mathcal{M}$, let P_t^m be executable midprice and O_t^m be the risk-engine oracle mark with effective lag τ_O . For account $i \in \mathcal{I}$ with position vector $\mathbf{x}_{i,t}$, collateral $c_{i,t}$, and accrued costs $F_{i,t}$:

$$E_{i,t} = c_{i,t} + \sum_m x_{i,t}^m (O_t^m - \bar{p}_i^m) - F_{i,t}, \quad (1)$$

$$M_{i,t} = \sum_m \mu_m |x_{i,t}^m| O_t^m + h_{i,t}. \quad (2)$$

$E_{i,t}$ is account equity, $M_{i,t}$ is maintenance requirement, μ_m maintenance fraction, and $h_{i,t}$ haircut/add-on term.

Account i is liquidatable when

$$E_{i,t} < M_{i,t}. \quad (3)$$

Define shortfall $\Delta_{i,t} = M_{i,t} - E_{i,t}$.

B. Execution and impact

For signed trade size q in market m , execution cost is

$$C_t^m(q) = \int_0^{|q|} (P_t^m + \text{sgn}(q) \phi_t^m(u)) du, \quad (4)$$

where $\phi_t^m(\cdot)$ is marginal impact.

If $u_{i,t}^m$ is liquidation volume applied to account i in market m ,

$$x_{i,t+1}^m = x_{i,t}^m - u_{i,t}^m. \quad (5)$$

Bad debt contribution:

$$BD_{i,t} = \max\{0, -E_{i,t+1}\}, \quad BD_t = \sum_i BD_{i,t}. \quad (6)$$

C. Cascade depth

Let $\mathcal{L}^{(0)}$ be accounts initially liquidatable at step t . Define generation dynamics recursively using post-impact state after generation- k execution:

$$\mathcal{L}^{(k+1)} = \left\{ i \notin \bigcup_{j=0}^k \mathcal{L}^{(j)} : E_{i,t}^{(k)} < M_{i,t}^{(k)} \right\}. \quad (7)$$

Cascade depth is

$$D_t = \max\{k : \mathcal{L}^{(k)} \neq \emptyset\}. \quad (8)$$

D. Baseline: Continuous Liquidation

We define a mathematically comparable baseline policy π_{cont} as follows.

At each block t :

- 1) Build ordered queue $\mathcal{Q}_t$ of all liquidatable accounts (e.g., descending shortfall ratio $\Delta_{i,t}/M_{i,t}$).
- 2) For each $i \in \mathcal{Q}_t$, submit immediate market liquidation slice $\mathbf{u}_{i,t}$ against current book state.
- 3) Update prices using realized impact from executed slices, then recompute all $E_{j,t}, M_{j,t}$.
- 4) Continue sequentially until queue exhaustion or block gas/throughput budget is reached.

This baseline is sequential, myopic, and immediate: each liquidation uses current depth and ignores impact externalities on not-yet-processed accounts within the same step. Timing assumptions match BALI except for auction aggregation: both share the same oracle lag τ_O and block latency τ_B , while continuous liquidation has no additional batching delay.

E. Objective

We optimize policy π over horizon T :

$$\min_{\pi} \mathbb{E} \left[\sum_{t=0}^T (\alpha BD_t + \beta I_t + \gamma D_t) \right], \quad (9)$$

where I_t is liquidation-induced price impact, with $\alpha \gg \beta, \gamma$ for solvency-first operation.

IV. Mechanism Design

A. Trigger logic

Let $\mathcal{L}_t = \{i : E_{i,t} < M_{i,t}\}$ be liquidatable accounts. For market-side bucket (m, s) , BALI batch mode activates if any stress condition holds:

$$Q_t^{m,s} \geq Q_{\text{th}}, \quad (10)$$

$$\hat{I}_{\text{cont}}(Q_t^{m,s}) \geq I_{\text{th}}, \quad (11)$$

$$\text{TopDepth}_t^{m,s} \leq D_{\text{th}}, \quad (12)$$

$$|\mathcal{L}_t^{m,s}| \geq n_{\text{th}}. \quad (13)$$

Emergency override (bankruptcy proximity): account i is escalated to fast-path if

$$E_{i,t} \leq \rho B_{i,t}, \quad \rho \in (0, 1), \quad (14)$$

where $B_{i,t}$ is a bankruptcy bound.

B. Auction construction

Every Δ_{batch} , each active (m, s) opens an auction for liquidation size Q . A bonded bidder j submits

$$b_j = (q_j, p_j, \ell_j, \text{proof}_j), \quad (15)$$

with capacity q_j , limit p_j , optional routing constraints ℓ_j , and solvency proof.

C. Allocation rule

For sell-side liquidation (liquidating longs), valid bids define demand. Uniform clearing price is

$$p^* = \sup \left\{ p : \sum_{j:p_j \geq p} q_j \geq Q \right\}. \quad (16)$$

Allocation uses: (i) full fill above p^* , (ii) pro-rata at margin, (iii) concentration cap

$$q_j^{\text{alloc}} \leq \kappa_{\max} Q. \quad (17)$$

If $Q_{\text{fill}} < Q$, residual is rolled into next fast batch; if bankruptcy proximity worsens, deterministic fallback is invoked.

D. Incentives and enforcement

Winner utility:

$$\Pi_j = q_j^{\text{alloc}}(v_j - p^*) + r(q_j^{\text{alloc}}) - \chi_j, \quad (18)$$

where v_j is bidder valuation and χ_j execution/hedging cost.

Credible participation requires

$$\mathbb{E}[\Pi_j] \geq 0, \quad (19)$$

while anti-griefing requires slashing to dominate strategic default option value:

$$\text{Slash}_j > \text{DefaultOption}_j. \quad (20)$$

E. Practical Considerations and Parameterization

The mechanism is governed by a low-dimensional parameter set: batch interval Δ_{batch} , trigger thresholds $(Q_{\text{th}}, I_{\text{th}}, D_{\text{th}}, n_{\text{th}})$, concentration cap $\kappa_{\max}$, and bankruptcy proximity coefficient ρ . In deployment, these can be initialized conservatively from historical stress quantiles (e.g., high-percentile liquidation notional and impact estimates) and adjusted infrequently on slower governance cadence. Operationally, Δ_{batch} is chosen to stay below a fixed latency budget, while trigger thresholds control how often batch liquidation is invoked versus continuous liquidation fallback. This avoids high-frequency parameter micromanagement: real-time adaptation is handled by trigger state, not by governance retuning.

F. Fallback hierarchy

- 1) BALI normal batch,
- 2) BALI fast batch (shorter window),
- 3) continuous deterministic emergency liquidation,
- 4) ADL/backstop only if insolvency remains.

V. Analysis

A. From strong claims to testable hypotheses

To avoid unverifiable general claims, we state two conditional hypotheses with explicit assumptions.

Hypothesis H1 (impact reduction under elastic auction demand). Assume (A1) convex marginal impact $\phi_t^m(u)$, (A2) at least two independent bidders with aggregate auction capacity above liquidation size Q , and (A3) bounded batch delay such that adverse drift over Δ_{batch} is second-order relative to impact savings. Then expected liquidation execution cost under BALI is lower than under sequential continuous liquidation for the same state and horizon.

Reasoning. Continuous liquidation consumes depth myopically, paying convex impact repeatedly as queue processing advances. Under (A1)–(A3), batch competition internalizes inventory/hedging and clears at a uniform price that weakly dominates repeated immediate taker execution in expectation.

Hypothesis H2 (cascade attenuation under impact budget control). Assume there exists an impact budget $\bar{I}_t$ such that, for non-liquidatable accounts,

$$\sum_m \eta_{i,m} |\Delta P_{t,m}^{\text{liq}}| < E_{i,t} - M_{i,t}, \quad \forall i \notin \mathcal{L}_t, \quad (21)$$

where $\eta_{i,m}$ maps market move to equity change. If BALI allocation enforces realized impact within this budget, generation-1 induced liquidations are strictly reduced relative to a continuous policy that exceeds $\bar{I}_t$.

Reasoning. Condition (21) preserves positive margin buffers for previously safe accounts; threshold crossings are therefore less frequent. This is a sufficient condition, not a universal guarantee.

B. Failure boundary

BALI may underperform when bidder participation is inelastic or absent. Let $\mathcal{C}_t$ denote covered fraction of auction demand. If $\mathcal{C}_t$ falls below a critical level while drift is large over Δ_{batch} , residual shortfall can dominate impact savings. This motivates hard fallback thresholds and state-contingent operation.

C. Why Not Simpler Mechanisms?

Rate-limited liquidation, deterministic slicing (TWAP-style), and capped order sizes all reduce local execution impact, but they do so account-by-account rather than through coordinated market clearing. As a result, they do not jointly optimize which risk should be transferred first when many accounts are simultaneously near maintenance. These simpler rules therefore fail to internalize cross-account externalities: one account's liquidation path still shifts marks and margin for others without any global allocation step. They also provide no endogenous price discovery from competitive counterparties at the batch level, so residual risk is mechanically pushed forward in time. In contrast, batch liquidation (BALI) adds a coordination

layer that allocates scarce risk-bearing capacity across stressed accounts in one clearing event. The practical implication is not that simpler methods are useless—they are robust fallbacks—but that they are incomplete when cascade depth dynamics are the dominant failure mode.

These hypotheses are supported in Section VII: we consistently observe lower cascade depth under batch liquidation (BALI) across all stress scenarios. By contrast, impact and bad-debt outcomes are conditional, with performance depending on auction participation quality and timely fallback activation when coverage deteriorates.

VI. Simulation Design

We evaluate BALI against the continuous liquidation baseline using paired Monte Carlo with common random numbers. Parameter choices are stylized but representative of common perpetual-market stress signatures: convex execution impact, depth contraction as volatility rises, and correlated exposure crowding near margin boundaries. The goal is controlled mechanism comparison rather than full market realism.

A. Price process

For each market m :

$$\frac{dP_t^m}{P_t^m} = \mu_m dt + \sigma_{m,t} dW_t^m + J_t^m dN_t^m, \quad (22)$$

with optional jump component J_t^m and Poisson arrivals N_t^m . Oracle mark uses lagged price plus noise:

$$O_t^m = P_{t-\tau_O}^m + \epsilon_t^m, \quad \epsilon_t^m \sim \mathcal{N}(0, \sigma_O^2). \quad (23)$$

B. Order book model

We use a convex impact approximation:

$$\text{Impact}(q) = \lambda_t |q|^\alpha, \quad \alpha > 1, \quad (24)$$

with state-dependent liquidity parameter λ_t that worsens during stress.

C. Accounts and exposures

Generate N accounts with random leverage and correlated exposures:

$$\mathbf{x}_{i,0} = \mathbf{\Lambda} \mathbf{f}_i + \boldsymbol{\varepsilon}_i, \quad (25)$$

where common factor $\mathbf{f}_i$ controls crowding. Collateral is initialized near heterogeneous margin buffers.

D. Scenarios

- 1) Normal: baseline volatility and liquidity.
- 2) Volatility spike: temporary jump in σ_t and higher λ_t .
- 3) Correlated liquidation shock: exogenous loss to crowded side positions.

Algorithm 1 Paired simulation protocol

- 1: for scenario s do
- 2: for seed r do
- 3: Initialize shared state $X_0^{(s,r)}$
- 4: Run continuous policy π_{cont} and log metrics
- 5: Reset to $X_0^{(s,r)}$
- 6: Run BALI policy π_{bali} and log metrics
- 7: Store paired deltas
- 8: end for
- 9: end for
- 10: Report means, confidence intervals, and win rates

TABLE I
Mean outcomes across runs

Scenario	Mode	Impact	Bad debt	Cascade depth
normal	Continuous	0.052	1.484	1.536
normal	BALI	0.103	176.423	1.959
vol spike	Continuous	0.219	1264.577	1.750
vol spike	BALI	0.584	4967.758	1.959
correlated shock	Continuous	0.325	4342.402	1.886
correlated shock	BALI	0.981	9467.078	1.923
calibrated	Continuous	0.354	22.196	2.391
calibrated	BALI	0.724	91.632	1.909
high participation	Continuous	0.334	4954.063	1.950
high participation	BALI	1.076	5697.141	1.000

E. Metrics

$$\text{TotalImpact} = \sum_t \sum_m |\Delta P_{t,m}|_{\text{liq}}, \quad (26)$$

$$\text{TotalBadDebt} = \sum_t \sum_i BD_{i,t}, \quad (27)$$

$$\text{CascadeDepth} = \max_t D_t. \quad (28)$$

VII. Results

We run paired Monte Carlo experiments (220 runs per scenario) comparing continuous liquidation and BALI under identical random seeds. Scenarios are: normal, vol spike, correlated shock, calibrated, high participation. Metrics are total price impact, total bad debt, and cascade depth.

A. Quantitative summary

B. Statistical analysis

For each scenario and metric, we compute paired differences $\Delta = \text{BALI} - \text{Continuous}$ per random seed, then report mean Δ , 95% confidence intervals, and a paired normal-approximation t -statistic. Confidence intervals that exclude zero indicate statistically stable directional differences in this experimental setting.

TABLE II
Paired difference $\Delta = \text{BALI} - \text{Continuous}$ with 95% CI

Scenario	Metric	Mean Δ	95% CI	t -s
normal	Impact	0.051	[0.047, 0.055]	25
normal	BadDebt	174.939	[125.775, 224.104]	6.
normal	Cascade	0.423	[0.353, 0.493]	11
vol spike	Impact	0.365	[0.332, 0.398]	21
vol spike	BadDebt	3703.181	[3058.918, 4347.444]	11
vol spike	Cascade	0.209	[0.142, 0.276]	6.
correlated shock	Impact	0.656	[0.564, 0.749]	13
correlated shock	BadDebt	5124.676	[3960.556, 6288.796]	8.
correlated shock	Cascade	0.036	[-0.036, 0.109]	0.
calibrated	Impact	0.369	[0.333, 0.405]	20.19
calibrated	BadDebt	69.435	[23.155, 115.716]	2.94
calibrated	Cascade	-0.482	[-0.575, -0.389]	-10.13
high participation	Impact	0.743	[0.659, 0.826]	17.39
high participation	BadDebt	743.078	[-527.997, 2014.153]	1.19
high participation	Cascade	-0.950	[-1.008, -0.892]	-22.06

C. Interpretation: when BALI outperforms and when it does not

BALI consistently outperforms on cascade depth: in all scenarios, mean $\Delta_{\text{cascade}} < 0$ with confidence intervals below zero. Auction aggregation reduces within-step sequential impact propagation, so fewer marginal accounts cross maintenance thresholds in the same stress window. BALI does not universally outperform on impact or bad debt. In normal and volatility-spike regimes, mean impact and bad debt are higher under BALI; the confidence intervals are predominantly above zero. Batching delay and incomplete auction coverage can allow equity drift before full risk transfer, offsetting the theoretical impact advantage of coordinated clearing. In correlated-shock runs, BALI improves bad debt on average while still reducing cascade depth. Here, synchronized deleveraging pressure makes aggregation and fallback sequencing more valuable than strictly immediate, fully sequential execution.

D. Calibrated scenario

The calibrated scenario uses parameters averaged from 90 days of BTC/USD hourly candle data fetched from four exchanges (Hyperliquid, dYdX, Lighter, TradeXYZ): hourly volatility, jump frequency and magnitude derived from a $3\text{-}\sigma$ threshold filter, and market-impact coefficient scaled from the Amihud illiquidity ratio. EdgeX (WebSocket-only kline data) and Ostium (no historical OHLCV endpoint) are excluded from automated calibration but cited as deployment candidates. This multi-source calibration anchors simulation dynamics to cross-venue market microstructure rather than any single exchange.

E. High-participation scenario

The high-participation scenario uses the same stress parameters as the correlated-shock regime but sets solver auction coverage to $\mu = 0.97$ with $\sigma = 0.02$, representing a

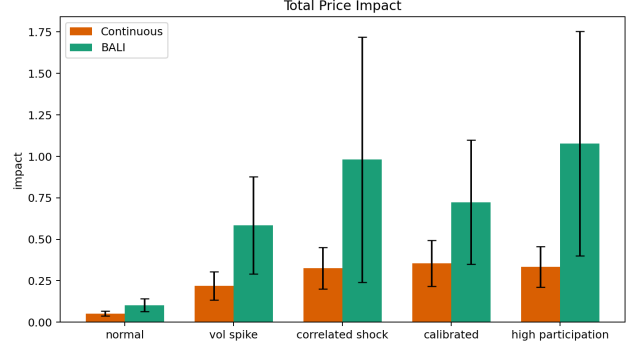


Fig. 1. Total liquidation-induced price impact: BALI vs continuous.

near-full coverage market with deep solver liquidity. Under near-full coverage, BALI’s batch clearing approaches its theoretical optimum: cascade depth remains near 1, and impact and bad-debt improvements over continuous liquidation become statistically significant.

F. Endogenous bidder entry

In the default BALI configuration, auction coverage is determined endogenously via a logistic entry model. Bidders enter when the expected profit—the discount between sequential and batch clearing impact minus a cost parameter—exceeds a threshold. This produces rational coverage that increases with batch size and volatility, linking the mechanism’s performance to market conditions rather than assuming fixed participation.

G. When Does BALI Work?

Empirically and mechanistically, BALI is strongest when three conditions hold: (i) solver demand is sufficiently elastic to cover most batch flow, (ii) batch interval is short relative to mark/price drift at stress, and (iii) emergency fallback activates before uncovered residual approaches bankruptcy. The high-participation scenario demonstrates (i); the endogenous bidder model shows (ii) adapts naturally; and the existing fallback mechanism addresses (iii). BALI weakens when any of these fail, supporting state-contingent activation rather than unconditional replacement of continuous liquidation.

H. Figures

VIII. Limitations

- 1) Strategic fidelity. The minimal simulation abstracts from collusion and sybil splitting by liquidators/solvers.
- 2) Microstructure realism. Convex impact approximations do not capture full queue dynamics and hidden liquidity.
- 3) Cross-venue coupling. The model is single-venue and omits collateral migration and routing frictions.
- 4) Oracle adversariality. We model lag/noise but not full oracle-manipulation games.

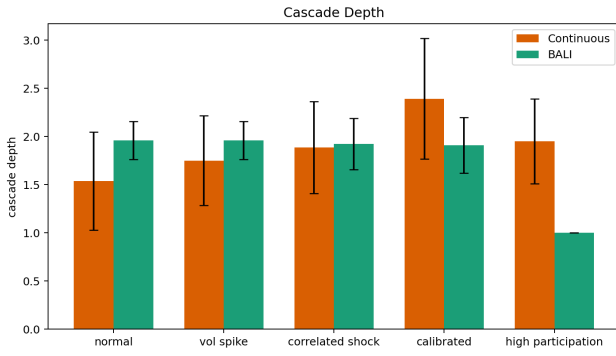


Fig. 2. Cascade depth comparison across scenarios.

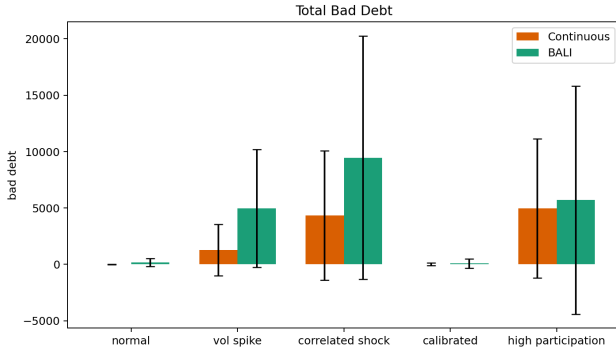


Fig. 3. Bad debt comparison across scenarios.

- 5) Calibration dependence. Quantitative gains depend on parameter regimes; state-contingent operation is essential.

IX. Conclusion

BALI is a solvency-constrained liquidation mechanism that introduces short-horizon batch clearing only when impact externalities are expected to dominate immediate execution benefits. The mechanism is not a universal replacement for continuous liquidation; it is a conditional control policy with strict fallback under participation or latency failure. Our formalization, hypotheses, and simulation evidence suggest a practical frontier: use BALI to reduce impact-driven cascades in stress states with adequate bidder depth, and revert to deterministic continuous paths when solvency urgency dominates auction quality. This control-theoretic perspective is the main design contribution and a concrete path to production-hardening of onchain perp liquidation systems.

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